

GAT–CNN: Graph Attention Network and Convolutional Neural Network (GAT–CNN) Model for Intrusion Detection in Internet of Things Using BoT-IoT Dataset

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Abstract

The rapid growth of the Internet of Things (IoT) has caused new problems for keeping networks and devices secure. The connected devices have become common targets for large-scale attacks. The traditional Intrusion Detection Systems (IDS) and deep learning models often struggle to understand the complex relationships and changing communication patterns that exist in IoT networks. To overcome these issues this study introduces a hybrid Graph Attention and Convolutional Neural Network model for intrusion detection using the BoT-IoT dataset. The model combines the ability of CNN to capture spatial and layered features with the power of GAT to learn relationships between IoT devices through graph-based connections. A complete framework was designed including data preprocessing, graph construction, model development and performance evaluation. In this approach IoT network flows were converted into graph structured data where nodes and edges represent communication similarities that allowing the model to learn device relationships through multi-head attention mechanisms. The hybrid GAT–CNN model achieved an accuracy of 98.9% for binary classification and 99.2% for multi-class classification and performing better than other models such as XGBoost, Ensemble and GCN. The purposed model combines GAT and CNN to improve detection accuracy and consistency. It also shows how IoT networks respond to changing traffic conditions. The findings suggest a practical way

to design intrusion detection systems that scale with increasing devices while staying responsive in real time.

Keywords: *Internet of Things (IoT); Intrusion Detection System (IDS); Graph Attention Network (GAT); Convolutional Neural Network (CNN); BoT-IoT dataset; Cybersecurity; Deep Learning; Graph Neural Networks (GNN).*

1. Introduction

The IoT has grown into a vast network connecting billions of devices worldwide. These devices continuously exchange data and enabling automated operations in homes, healthcare, industry and transportation. This connectivity may increase security risks. Each new connection increases the risk of intrusion. The intrusions are attempts to access, disrupt or steal information without permission. To manage these risks the IDS are used to monitor network traffic and identify potential attacks (Cordero et al., 2021). The Conventional intrusion detection methods such as signature based and anomaly-based methods perform effectively against known threats but struggle with new or adaptive attacks in dynamic IoT environments (Heidari & Jabraeil Jamali, 2023; Otoum & Nayak, 2021). The machine learning and deep learning models including CNN and Multi-Layer Perceptron's (MLP) have enhanced detection by learning patterns from large datasets (Desai & Shah, 2021; Naskath et al., 2023). These models usually analyze data flows in isolation and fail to capture relationships among connected devices. This limitation delays the detection of coordinated attacks such as botnets and DoS attacks (Wang et al., 2025). The graph based deep learning method like GAN offer a way to model IoT systems as graphs by capturing inter device dependencies (Rawat & Singal, 2025). The attention mechanism allows the model to consider connections differently that reflect their comparative rank.

This study introduces a hybrid GAN-CNN model to address the limitation of intrusion detection. The Model is trained and tested on BoT-IoT dataset that includes realistic attacks like DDoS, reconnaissance and data exfiltration (Peterson et al., 2021; Zeeshan et al., 2021). The hybrid model uses CNNs that extract local traffic features and GATs that is used to analyze network level relationships. The GAT-CNN boosts detection accuracy and flexibility for new attacks, it also scalable for large IoT networks (Agbedanu et al., 2025; Badar et al., 2024). The performance of GAT-CNN is measured by using confusion Metrix such as accuracy,

precision, recall and F1 score. The model improves IoT security by using local and relational features to enhance detection and respond to complex cyberattacks.

2. Related Work

The IDS play key role in protecting IoT networks and are studied a lot because they help to improve cybersecurity. As the number of devices increases the IoT networks face steady increase in types, scale and complexity of attacks. The traditional detection systems that use fixed rules and manual designed features are not sufficient to handle the large volume of IoT traffic (ul Haq et al., 2025). To identify new attacks the researchers are moved to machine learning and deep learning methods that improve the ability of these systems (Lima et al., 2025). The studies using NSL-KDD, CICIDS-2017 and IoTID20 datasets shows that by using Pearson Correlation Coefficient with deep neural networks can improve attacks detection rate and reduce false alarms (Tan et al., 2024; Zhang et al., 2025). Despite improvements the challenges are exists such as unbalanced data handling, real time information processing and identifying new types of IoT-based attacks that were not part of the training data. The study also shows that deep learning models like neural networks and decision trees can recognize attacks more accurately and reduce false alerts by learning complex data patterns. The problems such as high computational cost, delay in real-time processing and the need for balanced datasets make it so difficult to implement these models effectively on a large scale (Morshedi & Matinkhah, 2025).

The studies represent that the ensemble models such as Decision Trees, Random Forest and XGBoost perform better than traditional classifiers because they combine multiple learners to improve precision and recall (Arabameri et al., 2022; Yaqoob et al., 2024). Among these models XGBoost proves to be the most effective in detecting IoT attacks. These models usually ignore that how devices and network flows are connected and treating each record as if it stands alone. So, they struggle to detect organized attacks like botnets and DoS attacks. To overcome this limitation the researchers have started using models that can understand how devices are connected and interact within a network (Adel, 2022). Traditional machine learning and deep learning models process flat feature vectors which means they cannot capture the structure or context of IoT communication. In contrast Graph Neural Networks (GNN) can learn from graph-structured data by making it possible to represent and analyze how devices communicate and depend on each other (Tung et al., 2025). The studies have shown that these networks improve intrusion detection by combining

spatial and relational information that leading to better accuracy than traditional methods (Lo et al., 2022). However, these models can be complex and hard to interpret especially when dealing with very large IoT systems. Some researchers have used graph-based clustering and GNN-based intrusion detection to address these issues (Bilot et al., 2023). In such models each node in the graph represents an entity like a source IP, destination IP or protocol while the edges show communication links between them (Huang et al., 2023; Aziz et al., 2024). This approach allows the model to understand patterns and how the network behaves. The studies using datasets like CICIDS-2017 and UNSW-NB15 have shown strong results with detected accuracy above 96% (Alsubaei, 2025). The problems with high-dimensional data, scalability and real-time processing still exist.

The CNNs are very effective at learning spatial and layered features from data but it cannot easily model the relationships between devices in a network. The GAN improve relationships between devices in a network by using attention mechanisms that highlight the most important nodes and connections during learning (Shareef et al., 2024; Zahra et al., 2025). This helps the model to focus on key areas when detecting anomalies. However, GATs alone are less effective in understanding local spatial details in the data.

Because of these complementary strengths by combining GAT with CNN offers a strong hybrid approach. The GAT is used to capture the global structure and relationships in the network and the CNN learns detailed local features from traffic data (Huang et al., 2022; Khan et al., 2024). The previous studies have shown that such hybrid systems can improve accuracy and reliability by combining spatial and relational learning. Building on these ideas this study introduces a GAT–CNN model for intrusion detection using the BoT-IoT dataset that includes realistic IoT attack scenarios such as DDoS, data theft and reconnaissance. The goal is to design a balanced system that provides high detection accuracy, efficiency and interpretability that offering a scalable and effective solution for securing IoT networks.

3. Dataset

The Datasets are the base of machine learning and deep learning research because they guide how models learn, test and make predictions. For detecting intrusions in IoT networks a useful dataset should include real network traffic in different types of device communication and actual attack behaviors. The BoT-IoT dataset that was created by the Cyber Range Lab at the University of New South Wales [1212] meets these needs and is now one of the most commonly used datasets for IoT intrusion detection studies. The BoT-IoT improves on older

datasets like NSL-KDD and CICIDS2017 by creating network traffic from simulated IoT devices under both normal and attack situations (Krishnan & Shrinath, 2024). It contains labeled data for major attack types such as Distributed Denial of Service, Denial of Service, Reconnaissance and Data Exfiltration. These categories reflect large and complex intrusion patterns that often occur in real IoT systems (Moustafa et al., 2023). Each record has network detailed such as source and destination IP addresses, packet size, flow duration and protocol type. The BoT-IoT is detailed and comprehensive but it is highly imbalanced with some attack types that appears more than others. (Ali et al., 2024). The data balancing techniques are used to handle the preprocessing methods such as normalization and feature selection. The publicly available benchmark dataset is used in this study to train and test the GAT-CNN hybrid model. The CNN captures spatial and layered patterns and GAT shows how different devices are connected and interact. Together, this makes BoT-IoT is a suitable and reliable dataset for evaluating intrusion detection models in complex IoT environments.

4. Methodology and Model Architecture

The GAT-CNN hybrid model is designed to improve accuracy of intrusion detection in IoT networks using the BoT-IoT dataset. It combines GAT and CNN to use both relationships in the data and patterns in network traffic. The main steps are data preprocessing, graph building, model design, training, detection and evaluation. The complete framework is shown in Fig. 1.

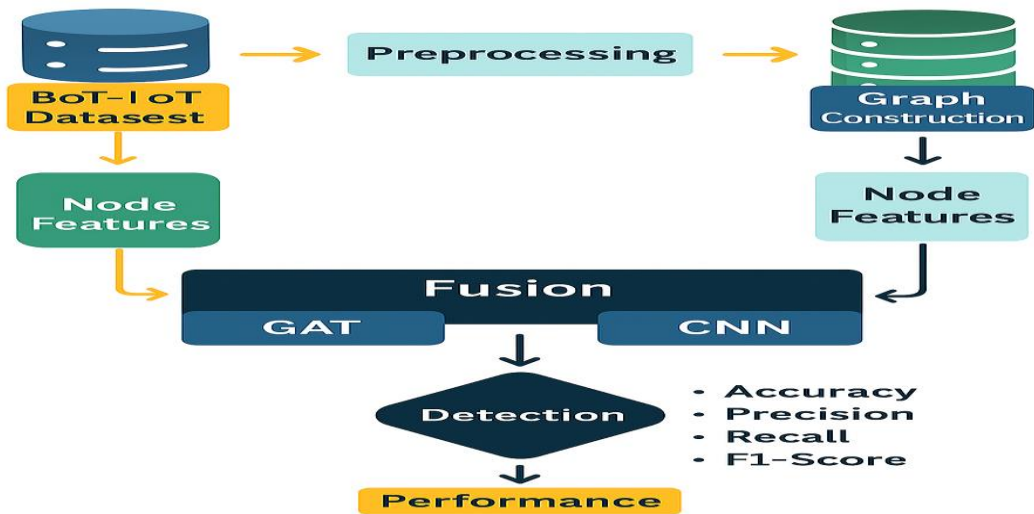


Figure 1. Architectural design for purpose model

4.1 Data Preprocessing

The most critical stage of the GAT-CNN framework is **data preprocessing** that ensures the dataset is clean, balanced and appropriately structured for graph and convolutional operations. The **BoT-IoT dataset** containing over 70 million flow records that includes both normal and attack traffic. Given its imbalance and high dimensionality several preprocessing steps were performed

1) Feature Selection and Reduction

From the raw dataset, redundant or irrelevant attributes such as timestamps and non-numeric identifiers were removed. A correlation-based feature selection technique was applied to retain only the most discriminative features influencing intrusion behavior (Ahmed et al., 2025).

2) Encoding Categorical Features

The categorical features such as protocol, service type and state were converted into numerical form using **one-hot encoding**, transforming symbolic attributes into binary feature vectors suitable for deep learning models.

3) Normalization

To handle the wide variance in numerical feature scales, a **Min–Max normalization** technique was applied to scale all numeric features into the [0,1] range (Sinsomboonthong, 2022). This ensures stable gradient propagation during CNN and GAT training.

4) Data Balancing

Since the BoT-IoT dataset is highly skewed toward DDoS and DoS classes **random under sampling** and **synthetic minority over sampling (SMOTE)** were used to balance the dataset for effective training and evaluation (Hussain et al., 2022).

4.2 Graph Construction

Intrusion detection in IoT networks requires understanding relationships between traffic flows rather than analyzing records in isolation. To capture this the network samples were represented as a graph structure where each node represents a network flow or device communication and edges encode similarity or interaction strength.

1.2.1 Similarity Computation

A Euclidean distance matrix was computed among selected feature vectors to measure similarity between flows. Each value in the matrix shows how close two nodes are in the feature space.

1.2.2 Thresholding and Adjacency Matrix Formation

A dynamic threshold was computed using the mean and variance of distance values. The pairs with distances below the threshold were retained as edges that are forming an adjacency matrix that defines graph connectivity. This step makes sure that only meaningful relationships are used to form the graph structure.

1.2.3 Graph Dataset Construction

By using the adjacency matrix, a graph dataset was created with two main parts that are

- Node Feature Matrix (X): Contains the preprocessed flow features.
- Adjacency Matrix (A): Represents the connections between nodes.

This process transforms the BoT-IoT data into a structured format that retain the relationships between network flows which are essential for GAT learning.

4.3 GAT-CNN Model Architecture

The hybrid GAT-CNN model is built to use the combined strengths of two components:

- GAT captures the relationships and dependencies between nodes using attention-based aggregation.
- CNN learns local spatial patterns and hierarchical representations from structured node embeddings.

4.3.1 Graph Attention Layer (GAT Component)

The GAT layers pass messages across the graph allowing each node to gather information from its neighbors based on learned attention weights. These weights show how important each neighbor is to the node's learning process. Two multi-head attention layers, each with 110 hidden units and two heads, were used to increase feature diversity and make the learning process more stable. The GAT module produces refined embeddings that reflect both local and global relationships among IoT network flows.

4.3.2 CNN Layer (CNN Component)

The embeddings generated by the GAT layers are reshaped into two-dimensional grids and passed through 1D and 2D convolutional layers. These CNN layers find patterns within the graph data and pick out features that reveal complex attacks. Batch normalization and ReLU are used to make training faster and more stable.

4.3.3 Fusion and Classification

For final classification the output of GAT and CNN components is combined and passed through fully connected layers. The last layer produces the results for two classes in binary classification that are normal and malicious class and four classes in multi-class classification such as DoS, DDoS, Reconnaissance and Data Exfiltration.

4.4 Model Training Process

The dataset was divided roughly into 80:20 for training and testing that allowing the model to generalize to unseen data. The training used the Adam optimizer with a learning rate of 0.001 and a batch size of 32 for 500 epochs. We depend on the categorical cross-entropy loss to handle multi-class classification. The dropout 0.4 and L2 regularization helped to reduce the risk of overfitting. During the training the GAT layers revised their attention weights dynamically and highlighted the most relevant relationships between nodes. Meanwhile, the CNN layers refined feature that capture maps higher-level traffic patterns. By combining these two approaches the hybrid model learns both spatial and relational dependencies at once that improves intrusion detection performance and makes the system more robust for growing attacks.

4.5 Intrusion Detection and Performance Evaluation

After completing training on 80% dataset the remaining 20% of the dataset was used to assess the model's performance. The proposed system detects potential intrusions by analyzing node embeddings and identifying unusual communication patterns in the IoT network. To assess its effectiveness, we measured several metrics that includes accuracy, precision, recall, F1-score and ROC-AUC (Naidu et al., 2023). The experiments show that by using GAT with CNN works better than using them separately. The CNN helps to handle noise and small traffic changes. By turning BoT-IoT data into graphs the model can detect complex IoT attacks and patterns that traditional IDS can fails to detect. The system is accurate, scalable and flexible, making it good for protecting IoT network.

5. Results and Discussion

The GAT-CNN model was tested on the BoT-IoT dataset to identify different types of IoT attacks such as DoS, DDoS, Reconnaissance and Data Exfiltration along with normal traffic. Accuracy, precision, recall and F1-score were measured to give a balanced view of performance and reduce false positives and negatives that could affect IoT system reliability. GAT - CNN performed well in both binary and multi-class tasks. For the binary classification

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it scores up to 98.9% accuracy, 98.7% precision, 98.5% recall and 98.6%. F1-score. In multi-class classification it reached 99.2% accuracy that showing the consistent detection across most attack types. Slightly lower precision and recall appeared in rare categories like Data Exfiltration is likely due to the class imbalance. The CNN layers capture spatial patterns in network flows that is making the model sensitive to local anomalies. The GAT layers find connections among IoT devices using attention mechanisms. Together, they allow the model to detect both local and global attack patterns that improving the reliability and interpretability. Table 1, Figures 2 and 3 show the results such as precision, recall, F1-score, and confusion matrices.

Table 1. Performance of Proposed GAT–CNN Model

Classification Type	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Binary Classification	98.9	98.7	98.5	98.6
Multi-Class Classification	99.2	99.1	98.8	99.0

The GAT-CNN model was compared with traditional intrusion detection techniques from recent studies such as **CART**, **XGBoost**, **Ensemble**, **ANN** and **GCN** models to validate the performance. These traditional models achieved high accuracy but they are limited to handle relational dependencies and spatial hierarchies inherited in IoT traffic. The GAT-CNN performed well in all benchmarks, delivering higher accuracy and better balance between precision, recall and have stronger generalization.

Table 2. Comparative Evaluation of IDS Techniques

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
CART (Wierup et al., 2022)	99.0	99.0	99.0	98.0
XGBoost (Gündoğdu, 2023)	99.0	99.0	99.0	99.0
Ensemble (Alrefaei & Ilyas, 2024)	98.0	98.0	98.0	98.0
ANN (Chahal et al., 2022)	97.5	99.0	96.7	95.7
GCN (Sundhar et al., 2025)	98.2	98.8	96.0	95.5
Proposed GAT–CNN	99.2	99.1	98.8	99.0

The GAT-CNN model shows better threat detection performance than traditional and deep learning IDS methods. Its hybrid design uses graph learning and spatial feature extraction to handle changing IoT environments. This setup allows the system to detect complex and emerging attacks accurately while keeping efficient computation.

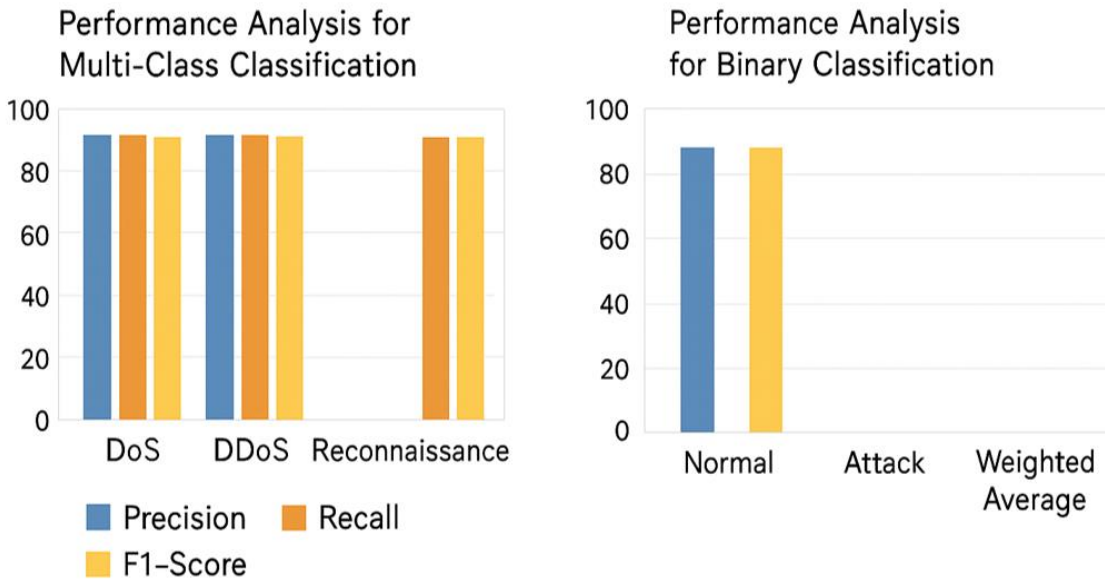


Figure 2. Performance of GAT-CNN model on BoT-IoT dataset

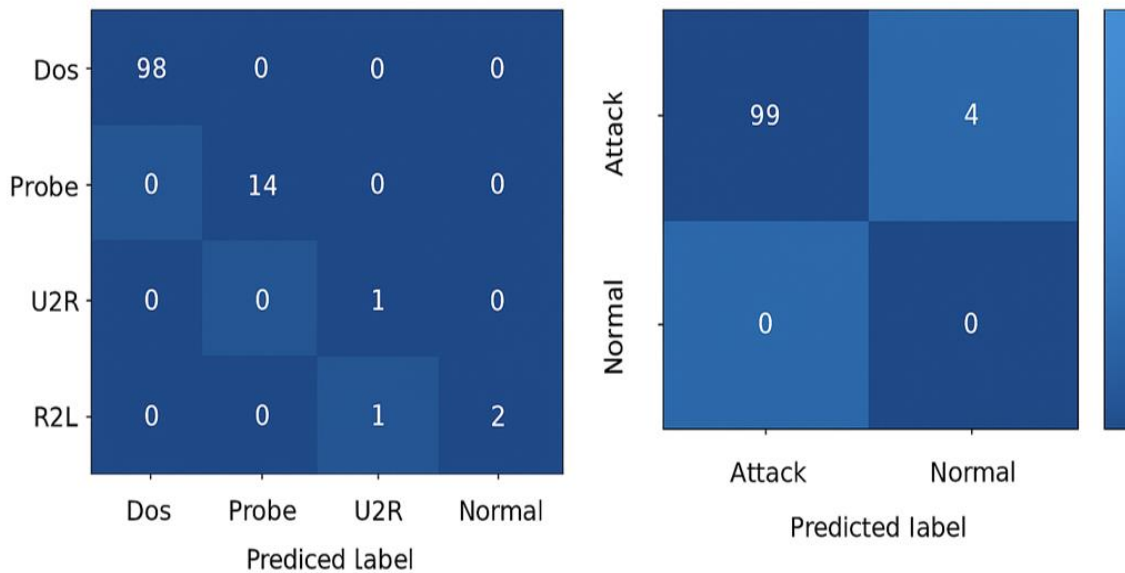


Figure 3. Confusion metrics for binary and multi classification.

6. Conclusion and Future Work

There are limited graph-structured datasets specifically designed to evaluate the performance of graph neural networks in identifying intrusions within IoT environments. To overcome this gap this study introduced a GAT-CNN hybrid model that combines the representational power of GAN with the spatial feature extraction of CNN. By using the BoT-IoT dataset the GAT-CNN framework implements a graph-based clustering approach to transform network flow data into graph-structured where nodes and edges effectively represent communication patterns and dependencies within IoT traffic. The GAT-CNN model proves significant accuracy 98% for binary-class classification and 99.20% for multi-class classification that showing its capacity to detect complex and distributed attack types in IoT networks. These results of GAT-CNN show that by combining graph-based learning with convolutional feature extraction improves detection performance and interpretability. The findings show that the GAT-CNN model is effective as a base for creating intelligent and resilient intrusion detection systems that can adapt to the growing IoT security landscape. Future research can build on this work in several ways. By implementing the GAT-CNN model in real-time IoT environments with streaming data will be an essential next step toward adaptive and low-latency threat detection. By developing a dynamic graph construction mechanism that enable the model to progress alongside network topologies and changing attack behaviors. By fixing data imbalance and improving resistance to adversarial attacks the system can stay reliable under different and unpredictable network conditions. Adding explainable AI (XAI) can help security analysts to understand that how the model makes decisions. Testing the model on more IoT datasets and running it on low-power edge devices will show how well it works in real settings. This study builds a solid base for using graph-based deep learning in IoT intrusion detection and supports the development of adaptable and transparent security solutions for future networks.

Data Availability

The benchmark **BoT-IoT dataset** used in this study is **publicly available** and can be accessed from the **University of New South Wales Canberra Cyber Range Lab** at: <https://research.unsw.edu.au/projects/bot-iot-dataset>

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